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Abstract

Action recognition is a crucial step in the process of human-computer interaction, especially in virtual reality, where it enables computers to understand and respond to human body movements. Previous work has utilized Graph Convolutional Neural Networks (GCNs) to extract skeletal features which already achieved excellent performance. However, it can only aggregate information between joints from normal skeleton graphs, making it difficult to effectively model complex interactions between joints, especially in scenarios where there is mutual dependence among multiple joints. To address this issue, we propose an Adaptive Hypergraph-Augmented Graph Convolutional Network (AH-GCN), which, while obtaining normal joints relationships, captures their high-order complex relationships by establishing hyperedges among multiple joints. Specifically, this method is based on normal graph convolution but uses an attention mechanism to learn both normal graph attention and hypergraph attention separately. The model samples hyperedge features from a learnable distribution and constructs a hypergraph based on the sampled hyperedges and the attention coefficients of the joints. Subsequently, the obtained normal attention and hypergraph attention are combined, enabling the model to focus on both normal and high-order motion information in body joints simultaneously. Our method has been tested on three large public datasets, NTU-RGB+D, NTU-RGB+D120 and Northwestern-UCLA, as well as on our own ethnic dance dataset, achieving state-of-the-art performance.

CCS Concepts

• Computing methodologies → Computer vision.

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Keywords

Human Skeleton, Action Recognition, Hypergraph, self-attention

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1 Introduction

Action recognition plays a key role in the field of human-computer interaction, especially skeleton-based recognition methods, which enable computers to understand human behavior and respond accordingly in virtual reality [27]. With the popularization of 3D depth cameras and the advancement of human pose estimation technology, human skeleton data has become much easier to obtain. Compared to traditional action recognition methods that use RGB video sequences [40], skeleton data not only captures more accurate motion information but also exhibits higher robustness to factors such as lighting, color, and occlusion. Early methods [14, 38] primarily relied on handcrafted features to accomplish the task of action recognition. However, with the progress of deep learning technology, methods based on recurrent neural networks (RNNs) [17, 29, 33] and convolutional neural networks (CNNs) [19, 23] have gradually become the mainstream. Despite this, these methods typically treat the human joints' coordinates as vector sequences or pseudo-images, which fails to adequately express the graph structure characteristics of skeleton data.

To address this issue, graph convolutional networks (GCNs) were proposed and successfully applied to action recognition [3, 30, 31, 44]. These networks are capable of better capturing the dependencies between joints, thereby improving the accuracy and robustness of action recognition. Yan et al. [44] was the first to use GCN to model skeleton data, proposing the spatio-temporal graph convolutional neural network (ST-GCN), which achieved significantly higher recognition accuracy than traditional methods. However, the manually specified skeleton topology is not applicable to all

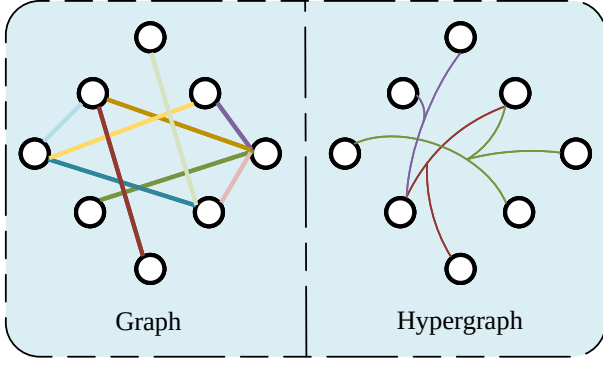


Figure 1: In a normal graph, an edge can only connect two joints, whereas in a hypergraph, a hyperedge may connect multiple joints simultaneously.

actions. Therefore, Chen et al. [3] proposed CTR-GCN, a method that dynamically generates variable topologies based on sample features, enabling the model to adaptively learn suitable topologies for different actions. The drawback of these methods is that the spatial topologies they use find it difficult to reflect the dependencies between multiple joints that are not physically connected. In complex scenarios, especially when there are mutual dependencies among multiple joints, the complex interactions between joints make it challenging for traditional graph neural network methods to perform comprehensive modeling, leading to difficulties in capturing high-order motion features of actions. Therefore, hypergraphs were introduced into action recognition [8, 41, 51]. Unlike normal graphs where an edge can only connect two joints, a hyperedge in a hypergraph can simultaneously connect multiple joints, making hypergraphs inherently superior to normal graphs in modeling complex joints relationships. Figure 1 intuitively illustrates this difference. However, in the method described in [51], the topology of the hyperedges is predefined manually, which limits the flexibility of the model.

We propose a novel Adaptive Hypergraph-Augmented Graph Convolutional Network (AHA-GCN). This network is designed to refine high-order motion features from action data, aiming to model complex joints relationships. The methodology involves the following steps: Firstly, a unique hyperedge construction mechanism is introduced, generating hyperedges through sampling and employing reparameterization skill [15] to optimize this sampling process. Subsequently, we innovatively focus on hyperedge-based feature extraction, treating the construction of the hypergraph as a clustering process, thereby enabling the model to dynamically and adaptively determine how the hypergraph is built. Following this, selective retention of the constructed hypergraph topology is performed, which is then integrated with topologies generated by normal attention mechanisms, ensuring that the model can capture both general and high-order motion features simultaneously. Finally, a transformation strategy is devised to convert the relationship from joints to hyperedges into joints-to-joints relationships, making the integration of these two distinct topological structures possible.

Our main contributions can be summarized as follows:

- We propose a hyperedge construction method that allows hyperedge features to be optimized during model training.
- We introduce AHA-GCN, which enables the model to adaptively learn hypergraph topologies based on normal graph convolutions, effectively capturing both normal and high-order motion features.
- We propose a strategy to transform the relationship between hyperedges and joints into relationships between joints.
- We conduct extensive experiments on multiple large datasets and our private dance dataset to demonstrate the superiority of our method.

2 Related Work

In this section, we will elaborate on the content that is related to our work.

2.1 Skeleton-based Action Recognition

In the early stages of research into skeleton-based action recognition, there was a prevalent reliance on handcrafted features to model the human skeleton and classify actions based on the trajectories of joints [14, 38]. For instance, Vemulapalli et al. [38] employed the relative geometric positions of joints to characterize skeleton data. While such approaches facilitated feature extraction tailored to specific requirements and were relatively straightforward to implement, they were limited in their ability to comprehensively model the nuanced information present in action data.

The advent of deep learning has seen a paradigm shift towards data-driven methodologies, which have increasingly supplanted traditional handcrafted feature approaches and become the actual standard within this domain. Recurrent Neural Networks (RNNs) [17, 29, 33] have been utilized to model skeleton data as sequential representations of joint coordinate vectors, with the subsequent sequences serving as inputs to the RNN models for predictive analysis. Conversely, Convolutional Neural Networks (CNNs) [17, 29, 33] have been applied to transform skeleton data into pseudo-image formats, capitalizing on the superior parallelization and ease of training afforded by CNN architectures.

Despite their advantages, both RNN and CNN methodologies fall short in fully capturing the intrinsic topological properties of skeleton data, which are intrinsically structured as graphs rather than sequences or two-dimensional grids. Recent advancements have seen the emergence of Graph Convolutional Networks (GCNs) [3, 30, 31, 44], notably including Spatio-Temporal Graph Convolutional Networks (ST-GCNs) [44], which directly model the graph structure of skeleton data. This approach obviates the necessity for manual design of part assignments or traversal rules, thus yielding superior performance compared to preceding methodologies. Furthermore, the introduction of self-attention mechanisms within frameworks such as 2s-AGCN [31] has augmented the capability to capture non-local dependencies, enhancing the overall representation learning capacity. To address the catastrophic forgetting issue in graph convolution, BlockGCN [50] utilizes graph distance to encode skeleton connections, efficiently capturing multi-relational information while maintaining the skeleton's topological structure.

2.2 Self-attention Mechanism

Self-attention mechanism can capture deep contextual information within the input sequence, effectively establishing long-range dependencies between tokens, which is difficult for convolutional neural networks due to their limited receptive fields. In visual tasks, self-attention mechanisms are widely utilized. For example, in the field of object detection, coordinate attention mechanisms combine channel information and positional information [11]. In semantic segmentation, criss-cross attention mechanisms obtain long-range dependencies while reducing memory usage [13]. In the field of action recognition, graph attention mechanisms have been widely applied [16, 33], where attention can be defined over joints [18], consistent with the definition of attention in this work, or it can be defined over edges [37].

2.3 Graph Convolutional Networks

Graph Convolutional Networks (GCNs) serve as an effective tool for processing non-Euclidean data and have garnered significant attention for their applications in graph-structured data. The development of GCNs primarily follows two streams: the spatial perspective and the spectral perspective [2, 10]. Methods from the spatial perspective directly apply convolutional filters to the joints and their neighbors in the graph, where these neighbors are typically extracted and normalized based on predefined rules [7, 25]. In contrast, methods from the spectral perspective leverage the eigenvalues and eigenvectors of the graph Laplacian matrix and perform convolution operations in the frequency domain through the graph Fourier transform. However, this approach relies on specific graph structural features (the Laplacian eigenbasis), making it applicable primarily to graphs with the same structure. Consequently, in practical applications, spatial methods are favored due to their simplicity and flexibility, especially when dealing with graphs where the number of joints or the connection patterns may vary. This study adopts the spatial perspective method for graph convolution.

2.4 Learning on Hypergraphs

The hypergraph breaks the traditional restriction that an edge can only connect two vertices in a graph structure, extending the concept of an edge into a hyperedge, which means that a hyperedge can connect multiple vertices. Complex relationships among entities can be connected by hyperedges, and thus can be represented using a hypergraph structure with higher-order semantics. In the field of action recognition, the complex relationships between skeleton joints are difficult to model comprehensively using normal graphs. Hao et al. [8] proposed a hypergraph neural network (Hyper-GNN) framework, where they divided skeleton data into three different input forms and used them as vertices to construct different types of hypergraphs to obtain higher-order feature representations of the skeleton. He et al. [9] proposed a dual-stream hypergraph convolutional network (SDHGCN), which adds a dual skeleton stream to a single skeleton stream to recognize interactive actions. T. Alsarhan et al. [1] proposed the disassembled hypergraph approach to focus on the most significant long-range edges within each hyperedge group, wherein the hyperedges are still manually constructed. In our work, we adopt the hypergraph construction method inspired by [49] to dynamically construct hypergraphs.

3 Proposed Method

In this section, we will first introduce the basics of hypergraphs. Subsequently, we will detail our proposed hyperedge construction method and the adaptive Hypergraph-Augmented Graph Convolution Network (AHA-GCN). In the AHA-GCN, we introduce the Hypergraph Attention Block (AHB) module. Finally, we will present a transformation strategy.

3.1 Preliminaries

Definition of Graph and Hypergraph The raw skeletal graph data consists of a time series of frames, with each frame containing vectors of body joint, which are generally represented as 2D or 3D coordinates. In traditional GCN based methods, a single frame of the human skeletal graph is represented as $G = (V, E)$ where $V = \{v_i | i = 1, \dots, n\}$ represents the set of joints, and $E = \{e_1, e_2, \dots, e_m\}$ represents the set of edges. Each $e_j (j = 1, 2, \dots, m)$ is a combination of two joints. However, as analyzed above, traditional GCN based methods assuming that each edge only links two vertices, the relationships between joints are more complex than pairwise in human actions. Thus, hypergraph is more effective than simple graph in describing complex relationships among multiple joints. Let $G = (V, E, W)$ be a hypergraph of human joints with n vertices and m hyperedges, where W is the weight of the hyperedges. The definition of V is the same as in a simple graph, and additionally, n and m represent the number of joints and the number of hyperedges, respectively. The sets of hyperedges can be defined as $E = \{e_1, e_2, \dots, e_m\}$. Where each $e_j (j = 1, 2, \dots, m)$ can be a combination of multiple joints.

In a normal graph, the features of the skeleton graph are represented by matrix $\hat{A}$. If there is an edge between two joints, the corresponding entry in the matrix is 1, otherwise it is 0. For the joint v_i , its degree matrix D_{v_i} represents the number of edges connected to it. The graph convolution formula can be expressed as:

$$X_{(l+1)} = \sigma \left(D_v^{-\frac{1}{2}} \hat{A} D_v^{-\frac{1}{2}} X_{(l)} W_{(l)} \right) \quad (1)$$

X_l is the input to l -th layer and W_l is the weight matrix. Hypergraph G can be represented by an incidence matrix $H \in \mathbb{R}^{n \times m}$ in general. The elements in the incidents matrix H of hypergraph can be defined as:

$$H = \begin{cases} h(v, e) = 1 & v \in V, e \in E, v \in e \\ h(v, e) = 0 & v \in V, e \in E, v \notin e \end{cases} \quad (2)$$

Definition of self-attention mechanism For an input sequence $X = \{x_1, x_2, \dots, x_n\}$, each x_i is first linearly transformed into three vectors: the query q_i , the key k_i , and the value v_i . Subsequently, the scores between each query vector q_i and key vectors k_j are calculated using the dot product formula [35]. These scores are then normalized into a weight matrix A_{ij} using the softmax function, ensuring that the sum of each row equals 1:

$$A_{i,j} = \vec{q}_i \cdot \vec{k}_j^\top \quad (3)$$

The final output at each position is computed as the weighted sum of all Values:

$$y_i = \sum_{j=1}^n A_{ij} v_j \quad (4)$$

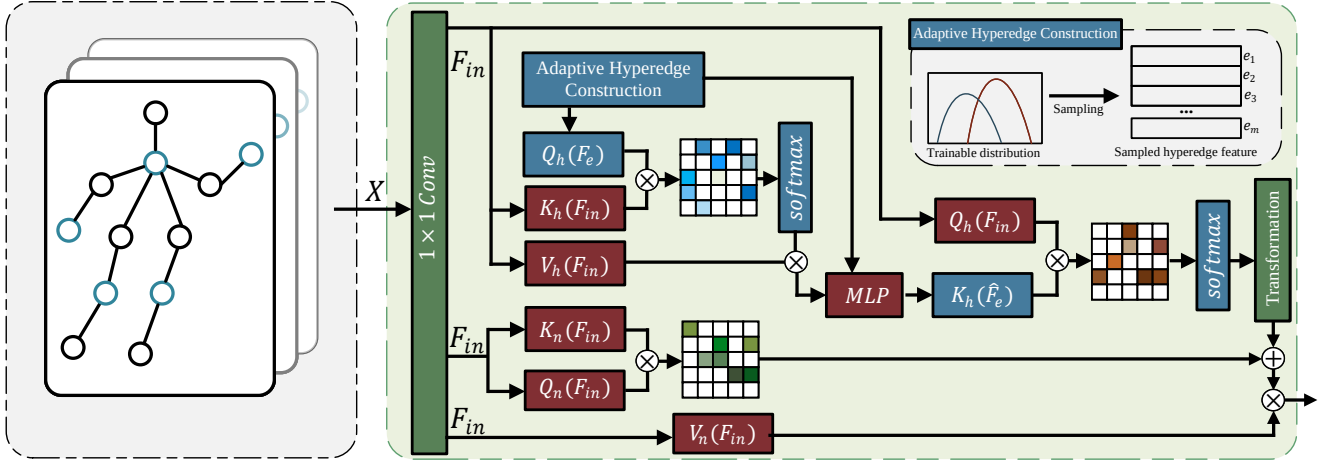


Figure 2: As shown in the figure, this is the overall structure of our method. The input data F_{in} has two paths, but they ultimately combine through element-wise summation. $\oplus$ represents element-wise summation and $\otimes$ represents matrix multiplication.

3.2 Adaptive Hyperedges Construction

Previous methods that involved hypergraphs for action recognition have been joint-centric, hence the topology of their hypergraphs is predefined, implying that the joints included in each hyperedge in H are fixed. However, different joints play varying roles in different human actions, and using a manually defined topological structure makes it difficult to effectively model the relationships between joints during different actions, thus limiting the representation power of the hypergraph. In this study, we innovatively adopt a hyperedge feature-centric approach. Based on a set of joints V and their feature matrix F_v , we aim to learn a hyperedge feature distribution $D(F_e|F_v)$, and use this distribution to dynamically construct the hypergraph topology H , thereby enabling dynamic updates to the topology of the hypergraph.

Assuming that for each skeleton graph $C \times T \times V$, where C is the dimension size, T is the temporal length, and V is the number of skeleton joints, it can be divided into m hyperedges, with each hyperedge containing k joints. For each hyperedge, we assume it follows a trainable Gaussian distribution $(F_e)_j \sim \mathcal{N}(\mu_j, \text{diag}(\sigma_j))$, where $\mu_j \in \mathbb{R}^{1 \times C}$, $\sigma_j \in \mathbb{R}^{1 \times C}$. Since sampling is a non-deterministic process and thus not differentiable, it cannot be updated through the loss function. Therefore, reparameterization skill [15] is used to make the entire sampling process differentiable. We first sample $G_j \in \mathbb{R}^{1 \times C}$ from a Gaussian distribution, and then F_e is obtained using the following formula:

$$(F_e)_j = \mu_j + \sigma_j \odot G_j \quad (5)$$

where $\odot$ represents the Hadamard multiplication. Both μ_j and σ_j are randomly initialized, and sampled m times to obtain the hyperedge matrix $F_e \in \mathbb{R}^{m \times C}$.

3.3 Adaptive Hypergraph-Augmented Graph Convolution Network

In equation 1, $D_v^{-\frac{1}{2}}$ represents the degree matrix and can be considered as a regularization term, which can be ignored. The equation

1 can be simplified to:

$$X_{l+1} = \sigma(\hat{A}X_lW_l) \quad (6)$$

here, $\hat{A}$ can be viewed as the attention matrix which will be discussed in detail. After multiplying X_l by the weight matrix W_l , the feature fusion is determined by $\hat{A}$. We hope that the model can not only extract features from normal graphs but also capture higher-order feature information through hypergraphs that is difficult to obtain from normal graphs. Thus in our method, $\hat{A}$ consists of two parts, namely the normal attention matrix $A_n \in \mathbb{R}^{B \times V \times V}$ and the hypergraph attention matrix $A_h \in \mathbb{R}^{B \times V \times V}$. Where B is the size of the batch. Hypergraph attention supplements normal attention to capture the higher-order correlations among joints that are difficult for normal graphs to learn. The overall architecture of our model is shown in Figure 2.

Normal Attention Matrix

$$A_n = \frac{Q_n(F_{in})\text{Re}(K_n(F_{in}))}{\sqrt{d_k}} \quad (7)$$

where the $\text{Re}()$ function represents the reshape operation. Through experiments, we did not convert $A_n \in \mathbb{R}^{B \times V \times V}$ into probability values but instead let it represent the correlation between joints.

Adaptive Hypergraph Attention Block In our method, the hypergraph serves as a supplement to normal graph attention, enhancing the connection strength between key joints that belong to the same hyperedge. As mentioned before, a trainable hyperedge matrix is obtained through sampling. However, since it is derived from sampling and unrelated to the features of the joints, it need to be refined further. We use the same attention calculation method as mentioned before to represent the correlation between hyperedges and each joint, including three trainable mapping functions Q_h, K_h, V_h . Here, we let the hyperedge query Q_h match the joint K_h ,

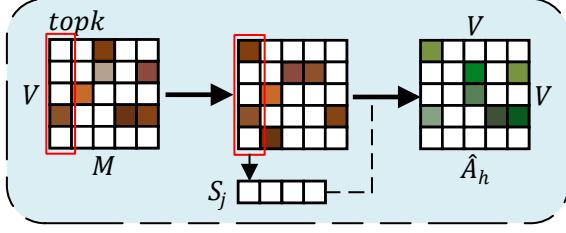


Figure 3: As shown in the figure, we retain the values of the top k joints in each column of A_h . Subsequently, the S_j array records the positions of these joints, and then guides (as indicated by the dashed lines) how each joint is added to $\hat{A}_h$.

which can be expressed as:

$$A_h = \text{Softmax} \left(\frac{Q_h(F_e) \text{Re}(K_h(F_{in}))}{\sqrt{d_k}} \right), \quad (8)$$

$$\hat{F}_e = A_h V_h(F_{in}).$$

A_h is the hyperedge attention matrix, a_{e_j, v_i} is the attention coefficient between hyperedge e_j and joint v_i . The hyperedge attention matrix determines which joints the hyperedge features are obtained from. Consequently, we can obtain the hyperedge feature matrix after integrating the joints' features $\hat{F}_e$. Then, F_e and $\hat{F}_e$ should be integrated:

$$\hat{F}_e = \text{MLP}(\text{concat}((F_e, \hat{F}_e))) \quad (9)$$

here, $\text{MLP}(\cdot)$ denotes the multi-layer perceptron, and $\text{concat}(\cdot)$ indicates concatenation. Through this approach, the randomly generated hyperedge features F_e are connected with the input joints' features F_{in} . Subsequently, the features of m hyperedges are obtained and they can be updated by backpropagating. We consider the construction of a hypergraph as a clustering process, with each hyperedge serving as a cluster center. The features of joints and cluster centers have now been obtained, and we simply need to assign each joint to the most relevant cluster. However, in a hypergraph, joints can belong to multiple hyperedges, so we still employ the attention mechanism to calculate the attention coefficients between joints and hyperedges, rather than assigning joints to a single hyperedge. Unlike using the hyperedge queries Q_h to match the joint keys K_h in the previous step, here we use the joint queries Q_h to match the hyperedge keys K_h . This process can be expressed as:

$$A_h = \text{Softmax} \left(\frac{Q_h(F_{in}) \text{Re}(K_h(\hat{F}_e))}{\sqrt{d_k}} \right) \quad (10)$$

here, $A_h \in \mathbb{R}^{B \times V \times m}$ is the hypergraph attention matrix, a_{v_i, e_j} is the attention coefficient between hyperedge v_i and joints e_j .

Transformation strategy For the input $F_{in} \in \mathbb{R}^{B \times C \times T \times V}$, after the above steps, we obtain noraml attention $A_n \in \mathbb{R}^{B \times V \times V}$ and hypergraph attention $A_h \in \mathbb{R}^{B \times V \times m}$. The latter serves as a supplement to the regular attention, and we need to add the two together. Therefore, it is necessary to transform the hypergraph attention into the same size as the noraml attention.

For each sample's hypergraph $A'_h \in \mathbb{R}^{V \times m}$, a'_{ij} is the element at the i -th row and j -th column. Firstly topk is used to retain the top

k joints related to each hyperedge, and other values will be set to 0. This process can be expressed as:

$$A'_{h_{ij}} = \begin{cases} a'_{ij}, & \text{if } a'_{ij} \in S_j \\ 0, & \text{if } a'_{ij} \notin S_j \end{cases} \quad (11)$$

here, S_j is the collection of the top k largest values in the j -th column of matrix A'_h . Then the following transformation is performed:

$$\hat{A}_{h_{ji}} = \hat{A}_{h_{ji}} + a_{ji}, \quad \forall i \in I_j \quad (12)$$

$\hat{A}_h$ is a $V \times V$ zero matrix. For the j -th column of A'_h , $I_j = \{i | a_{ij} \neq 0\}$ contains the row indices of all non-zero elements. We add this column to the corresponding row in $\hat{A}_h$. This transformation process is shown in Figure 3. This strategy converts the relationships between hyperedges and joints into relationships between joints and joints. Joints that play a significant role in multiple hyperedges are added to the noraml attention matrix after this transformation, as expressed by the following formula:

$$\hat{A} = A_n + \hat{A}_h \quad (13)$$

where A_n is the noraml attention matrix and $\hat{A}_h$ is the hypergraph attention matrix. Finally, after $\hat{A}$ multiplied by X_I , the result is passed to the subsequent temporal module for processing. The addition of both allows the model to focus on general motion features while also effectively capturing the special relationships between multiple joints in higher-order motion information, thereby enhancing the overall topological structure of the skeleton, leading to higher recognition accuracy for the model.

4 Experiments

In this section, we evaluate our proposed model on several public skeleton action recognition datasets, including NTU-RGB+D, NTU-RGB+D120, and Northwestern-UCLA. Additionally, we conducted experiments on our private ethnic dance dataset.

4.1 Datasets

NTU-RGB+D(NTU)[29] is currently the most widely used indoor-captured action recognition dataset. The dataset contains 56,880 action sequences covering 60 action categories. These sequences are performed by 40 volunteers aged between 10 and 35 years old. Each action sequence is executed by one or two subjects and is simultaneously captured by three Microsoft Kinect V2 depth sensors set at the same height but from different horizontal angles ($-45^\circ, 0^\circ, 45^\circ$). The dataset provides 3D joints locations for each frame, with 25 joints per subject in the skeleton sequences, and each video involves no more than two subjects. The original paper recommends two evaluation benchmarks: Cross-Subject (X-Sub) and Cross-View (X-View). For the X-Sub benchmark, the dataset is divided into a training set (40320 videos) and a test set (16560 videos), where the participants in the two subsets are different. For the X-View benchmark, the training set contains 37920 videos captured by cameras 2 and 3, while the test set includes 18960 videos captured by camera 1. We follow the convention and use Top-1 accuracy on this dataset.

NTU-RGB+D120(NTU120)[20] dataset is an extension of the NTU-RGB+D dataset, adding 57367 skeleton sequences and 60 additional action classes, making it the largest dataset containing 3D

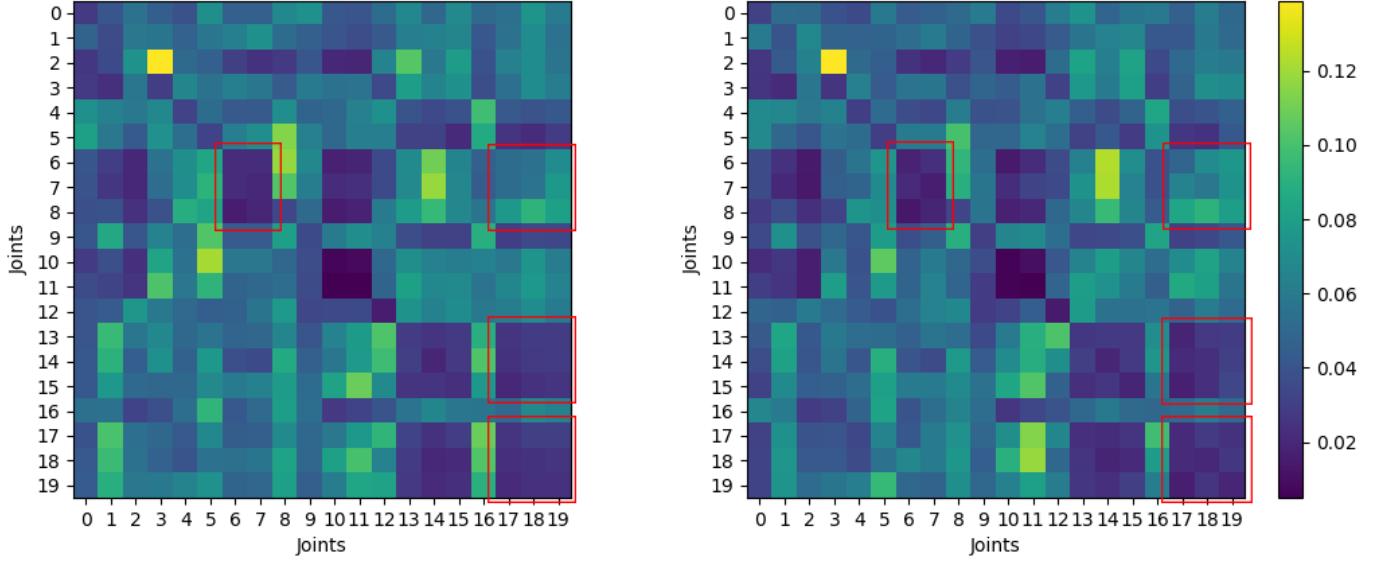


Figure 4: The attention matrix on the left is without the AHB module, while the one on the right includes the AHB module. It can be observed from the red markings that the model captures the relationships between joints more finely.

joints annotations. The NTU RGB+D 120 includes 32 setups, each representing a different location and background. The authors provided two benchmark evaluation schemes: Cross-Subject (X-Sub), where data from 106 subjects are divided into training and testing sets; and Cross-Setup (X-Set), which uses samples with even IDs for training and odd IDs for testing.

Northwestern-UCLA[39] dataset is a dataset designed for action recognition research, containing 1494 video sequences spanning 10 different action categories. These sequences are captured by three Kinect cameras from multiple viewpoints, which enhances the diversity of the dataset.

Private Ethnic Dance Dataset Our proprietary ethnic dance dataset uses eight high-speed infrared cameras to capture the three-dimensional motion trajectories of dancers. It includes 55 complete dances from 11 different ethnic groups, accumulating a total of 543,504 frames of RGB images and 9,840 motion data segments.

4.2 Details

Our model has 10 layers, with each layer having a dimensionality of 216. Initially, each layer reduces its dimensionality to 70 to decrease redundant parameters. The data flow, after being processed by our method, is up-dimensionalized back to 216, to be handled by subsequent modules. We utilize the PyTorch deep learning framework [26] and conduct experiments on a single RTX 4090 GPU, training for a total of 150 epochs. We use the standard cross-entropy loss function, and the model is optimized via stochastic gradient descent (SGD) with Nesterov momentum set at 0.90. The initial learning rate for all four datasets is set to 0.025, and it is reduced to one-tenth of its original value at epochs 110 and 120. We do not apply weight decay on the Northwestern-UCLA dataset, while for the other three datasets, it is set to 0.0004. For the NTU RGB+D and NTU RGB+D 120 datasets, as well as our private dataset, we set the batch size to

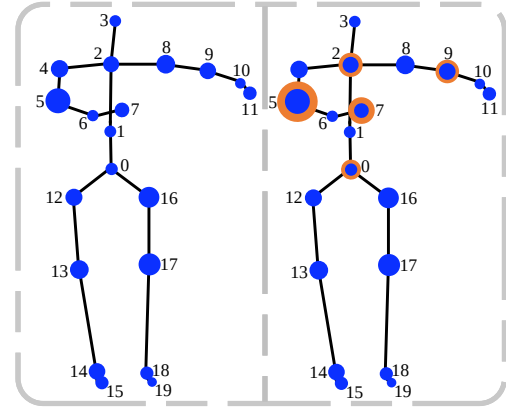


Figure 5: The left and right images are the skeleton graphs visualizing the node weights before and after adding the AHB module, respectively. In the right image, the AHB increases the weights of joints 0, 5, 7, 2, and 9 (indicated by orange). Intuitively, the elbow joints and fingers indeed play the most significant roles during the action of wearing clothes.

64 for the former two and to 16 for the latter, additionally resizing the samples to 64 frames. We adopt the data preprocessing code from [47]. For the Northwestern-UCLA dataset, the batch size is set to 16, and we follow the data preprocessing methods outlined in [3]. For a fair comparison, we use four streams of fusion results [3], considering joints, bone, and motion data stream forms.

Table 1: The effectiveness of AHB

Methods	Accuracy(%)
AHA-GCN wo/AHB(Js)	89.94
AHA-GCN(Js)	90.36 ^{↑0.42}

4.3 Ablation Study

We will validate the effectiveness of the method proposed in this paper through ablation studies conducted on the Cross-Sub protocol of the NTU-RGB+D dataset.

Adaptive Hypergraph Attention Block As mentioned earlier, our approach introduces hypergraph attention $\hat{A}_h$ as a supplement to the normal attention A_n . To verify the effectiveness of our hypergraph attention $\hat{A}_h$, we first utilize only the normal attention A_n and set it as the baseline. Table 1 demonstrates its performance. Under the same hyperparameter settings, the introduction of hypergraph attention significantly improves the accuracy. To more intuitively demonstrate the effect of Hypergraph Attention Block(AHB), we visualize the heatmaps of the attention matrices and skeleton weight maps with and without the AHB module on the Northwestern-UCLA dataset, as shown in Figure 4 and Figure 5. This heatmap pertains to the action "wearing clothes." As can be seen from the areas marked by red boxes, the AHB module enables the model to capture higher-dimensional information, thereby enhancing the differentiation of connection strengths between joints. This allows the model to integrate features from different joints in a more refined manner.

Table 2: Fixing k and m separately to obtain the optimal combination on NTU's X-Sub protocol.

(a) k fixed			(b) m fixed		
k	m	X-Sub (%)	m	k	X-Sub (%)
6	11	90.22 ^{↑0.28}	13	4	90.28 ^{↑0.34}
	12	90.10 ^{↑0.16}		5	89.90 ^{↓0.04}
	13	90.36 ^{↑0.42}		6	90.36 ^{↑0.42}
	14	90.10 ^{↑0.16}		7	90.15 ^{↑0.21}

Hypergraph Construction In the previous section, we proposed an adaptive hyperedge construction method. We assume that there are m hyperedges in the graph, with each hyperedge retaining the values of the top k joints, setting the rest to 0. These two hyperparameters significantly impact the performance of our method. Table 2 shows the experimental results on the NTU-RGB+D dataset. When the number of hyperedges m is set to 13 and k is set to 6, we achieved the highest accuracy of 90.36% on the X-Sub protocol of NTU-RGB+D, which is higher than the baseline of 89.94%. We believe that when m and k are too high, the excessive number of hyperedges leads to the model capturing a lot of redundant information, which cannot be effectively reduced by the topk operation. This diminishes the ability of the AHB module to strengthen the connections between key joints. Conversely, when m and k are too low, the model struggles to establish effective hyperedge relationships, which also weakens the effectiveness of the AHB module.

Table 3: Comparison with different strategy on NTU-RGB+D with X-Sub protocol

Strategy	Accuracy(%)
FC	89.87 ^{↓0.07}
Avg	90.12 ^{↑0.18}
selective	90.36 ^{↑0.42}

Table 4: Comparison on our private dataset

Type	Methods	Accuracy(%)
Transformer	ST-TR[28]	91.9
	DSTA[32]	92.1
GCN	InfoGCN[6]	92.6
	EfficientGCN[34]	92.7
	CTR-GCN[3]	93.2
	AHA-GCN(ours)	93.4

Table 5: Comparison with State-of-the-art on Northwestern-UCLA dataset

Type	Methods	Accuracy(%)
RNN	Lie Group[36]	74.2
	Ensemble TS-LSTM[17]	89.2
	2s-AGC-LSTM[33]	93.3
GCN,CNN	Va-CNN[43]	90.7
	SGN[47]	92.5
	Shift-GCN[5]	94.6
	DC-GCN+ADG[4]	95.3
	AHA-GCN(ours)	95.9

Transformation strategy Previously, we mentioned a transformation strategy used to convert the relationship between joints and hyperedges into the strength of connections between joints. This strategy was referred to as the selective transformation because we selectively decide which joints' connection strengths are retained (We believe that joints belonging to the same hyperedge have different connection strengths. We use the attention score of a joint belonging to the hyperedge as this connection strength). We also experimented with several other transformation methods, such as average transformation and fully connected transformation, and the results are shown in the Table 3. The average transformation involves averaging each column of A'_h and then broadcasting it to a size of $V \times V$. The fully connected transformation uses a full connected layer to transform A'_h to a size of $V \times V$. We believe that both the average transformation and the fully connected transformation introduce redundant information into the model, unable to focus on key joints as the selective transformation does, leading to a decrease in accuracy.

Table 6: Comparison with State-of-the-art on NTU dataset

Type	Methods	X-Sub(%)	X-View(%)
RNN	ST-LSTM[21]	69.2	77.7
	GCA-LSTM[22]	74.4	82.8
HyperGCN	DHGCN[41]	90.7	96.0
GCN	ST-GCN[44]	81.5	88.3
	2s-AGCN[31]	88.5	95.1
	FR-AGCN[12]	90.5	95.8
	Shift-GCN[5]	90.7	<u>96.5</u>
	MS-G3D[24]	91.5	96.2
	EfficientGCN[34]	91.7	95.7
	2M-STGCN[46]	90.8	96.2
	4s STF-Net[42]	91.1	<u>96.5</u>
	CTR-GCN[3]	92.4	96.8
	FRF-GCN[45]	91.3	<u>96.5</u>
	AHA-GCN(ours)	92.5	95.8

Table 7: Comparison with State-of-the-art on NTU120 dataset

Type	Methods	X-Sub(%)	X-Set(%)
RNN	ST-LSTM[21]	55.7	57.9
	GCA-LSTM[22]	61.2	63.3
HyperGCN	DHGCN[41]	86.0	87.9
GCN	ST-GCN[44]	70.7	73.2
	2s-AGCN[31]	82.5	84.2
	ST-TR[28]	82.7	84.7
	FRF-GCN[45]	87.1	88.4
	EfficientGCN[34]	88.3	89.1
	CTR-GCN[3]	88.9	<u>90.6</u>
	STFDN[48]	<u>89.3</u>	90.9
	AHA-GCN(ours)	89.4	<u>90.6</u>

4.4 Comparison with the State-of-the-arts

To validate the superiority of our proposed method, we compare it with other state-of-the-art skeleton-based action recognition methods on the NTU-RGB+D, NTU-RGB+D120, Northwestern-UCLA datasets, and our private dataset. The comparison results are presented in Table 4, Table 5, Table 6 and Table 7. The methods used for comparison include RNN-based methods, GCN-based methods and HyperGCN-based methods. Our model attained the highest performance in most of the comparisons.

5 Conclusion

In this paper, we propose the AHA-GCN, which includes several components: adaptive hyperedge construction, adaptive hypergraph building, and a transformation strategy. In our approach, the model can adaptively determine how to construct hyperedges, thereby avoiding the flexibility issues caused by manually defining hypergraphs in previous work [44]. We tested our method on multiple large datasets and our own private dance dataset, demonstrating

the superiority of our approach. In the future, we plan to explore other transformation strategies to better integrate the information learned from hypergraphs into normal graphs.

References

- [1] Tamam Alsarhan, Syed Sadaf Ali, Ayoub Alsarhan, Iyyakutti Iyappan Ganapathi, and Naoufel Werghi. 2024. Human Action Recognition with Multi-Level Granularity and Pair-Wise Hyper GCN. In *2024 IEEE 18th International Conference on Automatic Face and Gesture Recognition (FG)*. IEEE, 1–10.
- [2] Joan Bruna, Wojciech Zaremba, Arthur Szlam, and Yann LeCun. 2013. Spectral networks and locally connected networks on graphs. *arXiv preprint arXiv:1312.6203* (2013).
- [3] Yuxin Chen, Ziqi Zhang, Chunfeng Yuan, Bing Li, Ying Deng, and Weiming Hu. 2021. Channel-wise topology refinement graph convolution for skeleton-based action recognition. In *Proceedings of the IEEE/CVF international conference on computer vision*. 13359–13368.
- [4] Ke Cheng, Yifan Zhang, Congqi Cao, Lei Shi, Jian Cheng, and Hanqing Lu. 2020. Decoupling gcnn with dropgraph module for skeleton-based action recognition. In *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXIV* 16. Springer, 536–553.
- [5] Ke Cheng, Yifan Zhang, Xiangyu He, Weihao Chen, Jian Cheng, and Hanqing Lu. 2020. Skeleton-based action recognition with shift graph convolutional network. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 183–192.
- [6] Hyung-gun Chi, Myoung Hoon Ha, Seunggeun Chi, Sang Wan Lee, Qixing Huang, and Karthik Ramani. 2022. Infogcn: Representation learning for human skeleton-based action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 20186–20196.
- [7] Will Hamilton, Zhitao Ying, and Jure Leskovec. 2017. Inductive representation learning on large graphs. *Advances in neural information processing systems* 30 (2017).
- [8] Xiaoke Hao, Jie Li, Yingchun Guo, Tao Jiang, and Ming Yu. 2021. Hypergraph neural network for skeleton-based action recognition. *IEEE Transactions on Image Processing* 30 (2021), 2263–2275.
- [9] Changxiang He, Chen Xiao, Shuting Liu, Xiaofei Qin, Ying Zhao, and Xuedian Zhang. 2021. Single-skeleton and dual-skeleton hypergraph convolution neural networks for skeleton-based action recognition. In *Neural Information Processing: 28th International Conference, ICONIP 2021, Sanur, Bali, Indonesia, December 8–12, 2021, Proceedings, Part II* 28. Springer, 15–27.
- [10] Mikael Henaff, Joan Bruna, and Yann LeCun. 2015. Deep convolutional networks on graph-structured data. *arXiv preprint arXiv:1506.05163* (2015).
- [11] Qibin Hou, Daquan Zhou, and Jiashi Feng. 2021. Coordinate attention for efficient mobile network design. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 13713–13722.
- [12] Zesheng Hu, Zihao Pan, Qiang Wang, Lei Yu, and Shumin Fei. 2022. Forward-reverse adaptive graph convolutional networks for skeleton-based action recognition. *Neurocomputing* 492 (2022), 624–636.
- [13] Zilong Huang, Xinggang Wang, Lichao Huang, Chang Huang, Yunchao Wei, and Wenyu Liu. 2019. Ccnet: Criss-cross attention for semantic segmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*. 603–612.
- [14] Mohamed E Hussein, Marwan Torki, Mohammad A Gawayyed, and Motaz El-Saban. 2013. Human action recognition using a temporal hierarchy of covariance descriptors on 3d joint locations. In *Twenty-third international joint conference on artificial intelligence*.
- [15] Diederik P Kingma. 2014. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980* (2014).
- [16] Cuong Le and Xin Liu. 2023. Spatio-temporal Attention Graph Convolutions for Skeleton-based Action Recognition. In *Scandinavian Conference on Image Analysis*. Springer, 140–153.
- [17] Inwoong Lee, Doyoung Kim, Seoungyeon Kang, and Sanghoon Lee. 2017. Ensemble deep learning for skeleton-based action recognition using temporal sliding lstm networks. In *Proceedings of the IEEE international conference on computer vision*. 1012–1020.
- [18] John Boaz Lee, Ryan Rossi, and Xiangnan Kong. 2018. Graph classification using structural attention. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 1666–1674.
- [19] Hong Liu, Juanhui Tu, and Mengyuan Liu. 2017. Two-stream 3d convolutional neural network for skeleton-based action recognition. *arXiv preprint arXiv:1705.08106* (2017).
- [20] Jun Liu, Amir Shahroury, Mauricio Perez, Gang Wang, Ling-Yu Duan, and Alex C Kot. 2019. Ntu rgb+ d 120: A large-scale benchmark for 3d human activity understanding. *IEEE transactions on pattern analysis and machine intelligence* 42, 10 (2019), 2684–2701.
- [21] Jun Liu, Amir Shahroury, Dong Xu, and Gang Wang. 2016. Spatio-temporal lstm with trust gates for 3d human action recognition. In *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016*,

- Proceedings, Part III* 14. Springer, 816–833.
- [22] Jun Liu, Gang Wang, Ping Hu, Ling-Yu Duan, and Alex C Kot. 2017. Global context-aware attention lstm networks for 3d action recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 1647–1656.
 - [23] Mengyuan Liu, Hong Liu, and Chen Chen. 2017. Enhanced skeleton visualization for view invariant human action recognition. *Pattern Recognition* 68 (2017), 346–362.
 - [24] Ya-Qi Liu and Xiao-Yuan Wang. 2020. The analysis of driver’s behavioral tendency under different emotional states based on a Bayesian Network. *IEEE Transactions on Affective Computing* 14, 1 (2020), 165–177.
 - [25] Mathias Niepert, Mohamed Ahmed, and Konstantin Kutzkov. 2016. Learning convolutional neural networks for graphs. In *International conference on machine learning*. PMLR, 2014–2023.
 - [26] Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan, Edward Yang, Zachary DeVito, Zeming Lin, Alban Desmaison, Luca Antiga, and Adam Lerer. 2017. Automatic differentiation in pytorch. (2017).
 - [27] Patrick M Pilarski, Andrew Butcher, Michael Johanson, Matthew M Botvinick, Andrew Bolt, and Adam SR Parker. 2019. Learned human-agent decision-making, communication and joint action in a virtual reality environment. *arXiv preprint arXiv:1905.02691* (2019).
 - [28] Helei Qiu, Biao Hou, Bo Ren, and Xiaohua Zhang. 2022. Spatio-temporal tuples transformer for skeleton-based action recognition. *arXiv preprint arXiv:2201.02849* (2022).
 - [29] Amir Shahroury, Jun Liu, Tian-Tsong Ng, and Gang Wang. 2016. Ntu rgb+ d: A large scale dataset for 3d human activity analysis. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 1010–1019.
 - [30] Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. 2019. Skeleton-based action recognition with directed graph neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 7912–7921.
 - [31] Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. 2019. Two-stream adaptive graph convolutional networks for skeleton-based action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12026–12035.
 - [32] Lei Shi, Yifan Zhang, Jian Cheng, and Hanqing Lu. 2020. Decoupled spatial-temporal attention network for skeleton-based action-gesture recognition. In *Proceedings of the Asian conference on computer vision*.
 - [33] Chenyang Si, Wentao Chen, Wei Wang, Liang Wang, and Tieniu Tan. 2019. An attention enhanced graph convolutional lstm network for skeleton-based action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 1227–1236.
 - [34] Yi-Fan Song, Zhang Zhang, Caifeng Shan, and Liang Wang. 2022. Constructing stronger and faster baselines for skeleton-based action recognition. *IEEE transactions on pattern analysis and machine intelligence* 45, 2 (2022), 1474–1488.
 - [35] A Vaswani. 2017. Attention is all you need. *Advances in Neural Information Processing Systems* (2017).
 - [36] Vivek Veeriah, Naifan Zhuang, and Guo-Jun Qi. 2015. Differential recurrent neural networks for action recognition. In *Proceedings of the IEEE international conference on computer vision*. 4041–4049.
 - [37] Petar Velickovic, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lio, Yoshua Bengio, et al. 2017. Graph attention networks. *stat* 1050, 20 (2017), 10–48550.
 - [38] Raviteja Vemulapalli, Felipe Arrate, and Rama Chellappa. 2014. Human action recognition by representing 3d skeletons as points in a lie group. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 588–595.
 - [39] Jiang Wang, Xiaohan Nie, Yin Xia, Ying Wu, and Song-Chun Zhu. 2014. Cross-view action modeling, learning and recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2649–2656.
 - [40] Limin Wang, Yuanjun Xiong, Zhe Wang, Yu Qiao, Dahua Lin, Xiaoou Tang, and Luc Van Gool. 2016. Temporal segment networks: Towards good practices for deep action recognition. In *European conference on computer vision*. Springer, 20–36.
 - [41] Jinfeng Wei, Yunxin Wang, Mengli Guo, Pei Lv, Xiaoshan Yang, and Mingliang Xu. 2021. Dynamic hypergraph convolutional networks for skeleton-based action recognition. *arXiv preprint arXiv:2112.10570* (2021).
 - [42] Liyu Wu, Can Zhang, and Yuexian Zou. 2023. SpatioTemporal focus for skeleton-based action recognition. *Pattern Recognition* 136 (2023), 109231.
 - [43] Kailin Xu, Fanfan Ye, Qiaoyong Zhong, and Di Xie. 2022. Topology-aware convolutional neural network for efficient skeleton-based action recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 2866–2874.
 - [44] Sijie Yan, Yuanjun Xiong, and Dahua Lin. 2018. Spatial temporal graph convolutional networks for skeleton-based action recognition. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 32.
 - [45] Xiao Yun, Chenglong Xu, Kevin Riou, Kaiwen Dong, Yanjing Sun, Song Li, Kevin Subrin, and Patrick Le Callet. 2024. Behavioral Recognition of Skeletal Data Based on Targeted Dual Fusion Strategy. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 6917–6925.
 - [46] Haiping Zhang, Xu Liu, Dongjin Yu, Liming Guan, Dongjing Wang, Conghao Ma, and Zepeng Hu. 2023. Skeleton-based action recognition with multi-stream, multi-scale dilated spatial-temporal graph convolution network. *Applied Intelligence* 53, 14 (2023), 17629–17643.
 - [47] Pengfei Zhang, Cuiling Lan, Wenjun Zeng, Junliang Xing, Jianru Xue, and Nan-ning Zheng. 2020. Semantics-guided neural networks for efficient skeleton-based human action recognition. In *proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 1112–1121.
 - [48] Zhifu Zhao, Ziwei Chen, Jianan Li, Xiaotian Wang, Xuemei Xie, Lei Huang, Wanxin Zhang, and Guangming Shi. 2024. Glimpse and zoom: Spatio-temporal focused dynamic network for skeleton-based action recognition. *IEEE Transactions on Circuits and Systems for Video Technology* (2024).
 - [49] Peng Zhou, Zongqian Wu, Xiangxiang Zeng, Guoqiu Wen, Junbo Ma, and Xiaofeng Zhu. 2023. Totally dynamic hypergraph neural network. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*. 2476–2483.
 - [50] Yuxuan Zhou, Xudong Yan, Zhi-Qi Cheng, Yan Yan, Qi Dai, and Xian-Sheng Hua. 2024. BlockGCN: Redefine Topology Awareness for Skeleton-Based Action Recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2049–2058.
 - [51] Yiran Zhu, Guangji Huang, Xing Xu, Yanli Ji, and Fumin Shen. 2022. Selective hypergraph convolutional networks for skeleton-based action recognition. In *Proceedings of the 2022 international conference on multimedia retrieval*. 518–526.